

MATH4069 Notes

Measure Theory and Fourier Transform (Advanced)

Measurable Sets

Definition 1: σ -algebra

$\mathcal{A}$ is said to be a σ -algebra of X , are a collection of subsets of X that satisfy:

- Empty set: $\emptyset \in \mathcal{A}$
- Closure under compliments: $A \in \mathcal{A} \Rightarrow A^c \in \mathcal{A}$
- Closed under unions: $A_j \in \mathcal{A} \Rightarrow \bigcup_{j \in \mathbb{N}} A_j \in \mathcal{A}$

Example: A set $\mathcal{A}$ is a σ -algebra, show that $\mathcal{A}_S := \{A \cap S : A \in \mathcal{A}\}$ is a σ -algebra of S .

1. $\emptyset \in \mathcal{A}_S$ since $\emptyset \in \mathcal{A} \Rightarrow S \cap \emptyset \in \mathcal{A}_S$
2. Let $B \in \mathcal{A}_S \Rightarrow B := A \cap S \Rightarrow B^c := S \setminus B = A^c \cap S \in \mathcal{A}_S : A^c \in \mathcal{A}$
3. Let $\{B_j\} \in \mathcal{A}_S \Rightarrow \forall j \in \mathbb{N} B_j := A_j \cap S$ then, $\bigcup B_j = \bigcup A_j \cap S = S \cap \bigcup A_j \in \mathcal{A}_S : \bigcup A_j \in \mathcal{A}$

Definition 2: Measure

A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a measure if:

- Empty set: $\mu(\emptyset) = 0$
- Countable additivity: $A_j \in \mathcal{A}$ are disjoint, then $\mu(\bigcup_{j \in \mathbb{N}} A_j) = \sum_{j=1}^{\infty} \mu(A_j)$

Example: A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is a measure. Show that $\mu_{S(A)} := \mu(A \cap S)$ is also a measure.

1. $\mu_S(\emptyset) = \mu(\emptyset \cap S) = \mu(\emptyset) = 0$
2. $\mu_S(\bigcup A_j) = \mu((\bigcup A_j) \cap S) = \mu(\bigcup (A_j \cap S)) = \sum \mu(A_j \cap S) = \sum \mu_S(A_j)$

Definition 3: Outer Measure

A function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ is called an outer measure if:

- Empty set: $\mu^*(\emptyset) = 0$
- Countable sub-additivity: $A, A_j \in \mathcal{A}$, with $A \subseteq \bigcup_{j=0}^{\infty} A_j$ then $\mu^*(A) \leq \sum_{j=1}^{\infty} \mu^*(A_j)$

Definition 4: Lebesgue Outer Measure

The function m_N^* is called the Lebesgue Outer Measure and is defined on $\mathbb{R}^N$ as:

$$m_N^* := \inf \left\{ \sum_{k=0}^{\infty} \text{vol}(R_k) : A \subseteq \bigcup_{k=0}^{\infty} R_k, R_k \text{ open rectangles} \right\}$$

An open rectangle $R_k \in \mathbb{R}^N$ can be defined as $R_k = (a_0, b_0) \times \dots \times (a_N, b_N)$, this means that the volume is defined as

$$\text{vol}(R_k) = (b_1 - a_1) \dots (b_N - a_N)$$

Definition 5: Measurable Sets

A set $A \subseteq X$ is said to be measurable, with respect to a measure (or outer measure) μ (or μ^*) if for a generic set $S \subseteq X$ it satisfies

$$\mu^*(S) = \mu^*(S \cap A) + \mu^*(S \cap A^c)$$

Definition 6: Lebesgue Measure

The Lebesgue Measure, m_N^* on $\mathbb{R}^N$ is defined as Since the Lebesgue Measure is a Borel Measure, all

open and closed sets are automatically measurable. All countable unions/intersections of closed and open sets are measurable.

Definition 7: Borel Sets

A Set B is said to be Borel if for the collection of all open sets in X , C , B is generated by $\mathcal{A}(C)$. In general, we say that the following sets are Borel:

- all open sets (definition)
- all closed sets (compliment of open sets)
- all countable unions/intersections of open/closed sets (closure under unions)
- all countable unions/intersections of Borel sets

Measurable Functions

Definition 1: Measurable Functions

A function $f : X \rightarrow \mathbb{K}^N$ is said to be measurable if for all open sets $U \subseteq \mathbb{K}^N$, $f^{-1}[U] \in \mathcal{A}$.

Remark:

- f is measurable if and only if $f^{-1}[B] \in \mathcal{A}$ for every Borel set $B \in \mathbb{K}^N$
- $f : X \rightarrow \mathbb{R}$ is measurable if and only if $f^{-1}((\alpha, \infty)) \in \mathcal{A}$ for every rational $\alpha \in \mathbb{R}$
- All continuous functions are Borel Measurable

Definition 2: Indicator Functions

An indicator function is defined with a set $A \subseteq X$ with

$$1_A : X \rightarrow \mathbb{R}(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Remark: The indicator function is measurable if and only if the set A is measurable.

Definition 3: Simple Functions

A simple function can be defined as a weighted sum of indicator functions, where $\{A_k\} \in X$ are disjoint and $\alpha_k \in \mathbb{R}$

$$f(x) = \sum_{k=1}^m \alpha_k 1_{A_k}(x)$$

Definition 4: Positive Measurable Functions

For every measurable function $f : X \rightarrow [0, \infty]$ there exists a sequence of simple measurable functions $\varphi_n : X \rightarrow [0, \infty)$ such that $0 \leq \varphi_n \leq \varphi_{n+1} \leq f$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} \varphi_n(x) = f(x)$ for all x in X

Remark: We can utilise the Monotone Convergence Theorem to prove that a function is integrable.

Example: For a measurable function $g : X \rightarrow [0, \infty]$, $\nu(A) := \int_A g d\mu := \int_X g 1_A d\mu$, prove that $\int_X f d\nu = \int_X f g d\mu$ by starting with simple measurable, then positive and finally real-valued functions.

1. If f is simple then $f := \sum_k \alpha_k 1_{A_k}$ and $\int_X f d\nu = \sum_k \alpha_k \nu(A_k) = \sum_k \alpha_k \int_{A_k} g d\mu = \int_X \sum_k \alpha_k 1_{A_k} g d\mu = \int_X f g d\mu$
2. If f is positive, then $\exists \varphi_n$ simple such that $\varphi_n \rightarrow f$ then $\int_X f d\nu = \lim_{n \rightarrow \infty} \int_X \varphi_n g d\mu = \int_X \lim_{n \rightarrow \infty} \varphi_n g d\mu = \int_X f g d\mu$
3. If f is real valued, then $f = f^+ - f^-$ where both f^+ and f^- are positive functions, then $\int_X f d\nu = \int_X f^+ d\nu - \int_X f^- d\nu = \int_X f^+ g d\mu - \int_X f^- g d\mu = \int_X f g d\mu$

Integration

Definition 1: Integration of Simple Functions

If $\varphi : X \rightarrow [0, \infty) = \sum_{k=0}^n \alpha_k A_k$ is a positive, simple, measurable function then we can define the integral of φ as:

$$\int_X \varphi(x) d\mu = \sum_{k=0}^n \alpha_k \mu(A_k)$$

Remark: The following are properties of the Integration of Measurable Functions:

- Addition: $\int_X \varphi + \psi d\mu = \int_X \varphi d\mu + \int_X \psi d\mu$
- Projection: $\int_X \alpha \varphi d\mu = \alpha \int_X \varphi d\mu$
- Monotonicity: for $0 \leq \varphi \leq \psi$, $0 \leq \int_X \varphi d\mu \leq \int_X \psi d\mu$

Definition 2: Monotone Convergence Theorem

For a sequence of functions $f_n : X \rightarrow [0, \infty]$, with $0 \leq f_n \leq f_{n+1}$, $\forall n \in \mathbb{N}$ then,

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X \lim_{n \rightarrow \infty} f_n d\mu = \int_X f d\mu$$

This theorem provides us the necessary conditions to pass a limit through an integral.

Definition 3: Fatou's Lemma

Let $f_n : x \rightarrow [0, \infty]$ be an arbitrary measurable function, then

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu$$

Definition 4: Integrability

A function $f : X \rightarrow \mathbb{K}$ is said to be integrable, or $f \in \mathcal{L}^1(X)$ if

$$\int_X |f| d\mu < \infty$$

Definition 5: Dominated Convergence Theorem

On a measure space $(X, \mathcal{A}, \mu)$, a series of functions $f_n : X \rightarrow \mathbb{K}$ are measurable functions such that $f_n \rightarrow f$ and if there exists a $g \in \mathcal{L}^1(X)$ such that $|f_n| \leq g$ for all $n \in \mathbb{N}$, then we can say

$$\lim_{n \rightarrow \infty} \int_x f_n d\mu = \int_x \lim_{n \rightarrow \infty} f_n d\mu = \int_x f d\mu$$

Definition 6: Continuity of Parameter Integrals

If the function, $f : X \times Y \rightarrow \mathbb{K}$, satisfies the following conditions:

- $x \rightarrow f(x, y)$ is measurable for all y
- $y \rightarrow f(x, y)$ is continuous for all x
- $\exists g \in L^1(X; \mathbb{R})$ such that $|f(x, y)| \leq g(x)$

Then we can say that

$$F(y) := \int_X f(x, y) d\mu$$

is continuous at y

Definition 7: Differentiation of Parameter Integrals

If the function, $f : X \times Y \rightarrow \mathbb{K}$, satisfies the following conditions:

- $x \rightarrow f(x, y)$ is integrable
- $\frac{\delta f}{\delta y}$ exists and is continuous
- $\exists g \in L^1(X; \mathbb{R})$ such that $\left| \frac{\delta f}{\delta y} \right| \leq g(x)$

Then for a function $F(y) := \int_X f(x, y) d\mu$, we can define the derivative of F as:

$$F'(y) := \int_X \frac{\delta f}{\delta y} d\mu$$

Definition 8: p-Lebesgue Measure

We can define the function $\|\cdot\|_p$ as $\|u\|_p := \left(\int_X |u|^p d\mu \right)^{\frac{1}{p}}$ and the L-p space as follows

$$\mathcal{L}^p(X) = \{f : X \rightarrow \mathbb{K} \text{ measurable} : \|u\|_p < \infty\}$$

We can then define a space $L^p := \mathcal{L}^p \setminus \mathcal{N}(X)$ where $\mathcal{N}(X)$ is the set of zero almost everywhere functions on X.

Definition 9: Inequalities of Integrals

1. Holder's Inequality: $|\int_X fg d\mu| \leq \int_X |fg| d\mu \leq \|f\|_p \|g\|_q$ s.t. $\left(\frac{1}{p}\right) + \left(\frac{1}{q}\right) = 1$
2. Minkowski's Inequality (Triangle Inequality): $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Definition 10: Tonelli's Theorem

Tonelli's Theorem applies to positive function $f : X \times Y \rightarrow [0, \infty]$ and states that if:

- $x \rightarrow f$ is measurable
- $y \rightarrow f$ is measurable
- $x \rightarrow \int_Y f dy$ is measurable
- $y \rightarrow \int_X f dx$ is measurable

Then we can say that $\int_X \int_Y f dy dx = \int_Y \int_X f dx dy$

Definition 11: Fubini's Theorem

Fubini's Theorem applies to any function $f : X \times Y \rightarrow \mathbb{K}$ and states that if:

- $\int_X \int_Y |f| dy dx$ or $\int_Y \int_X |f| dx dy$ are finite
- $x \rightarrow f$ is measurable
- $y \rightarrow f$ is measurable
- $x \rightarrow \int_Y f dy$ is measurable
- $y \rightarrow \int_X f dx$ is measurable

Then we can say that $\int_X \int_Y f dy dx = \int_Y \int_X f dx dy$

Convolutions and Fourier Transforms

Definition 1: Convolution

If $f, g : \mathbb{R}^N \rightarrow \mathbb{K}$ are measurable functions, then we can define the convolution of $f * g$ as

$$(f * g)(x) = \int_{\mathbb{R}^N} f(x-y)g(y)dy$$

Definition 2: Young's Inequality

For two functions, $f \in L^1$ and $g \in L^p$ and $f * g \in L^p$, then

$$\|f * g\|_p \leq \|f\|_1 \|g\|_p$$

Moreover, if $f \in L^p$ and $g \in L^q$ and $\left(\frac{1}{p}\right) + \left(\frac{1}{q}\right) = 1$, then

$$\|f * g\|_\infty \leq \|f\|_p \|g\|_q$$

Definition 3: Approximate Identity

A function $\varphi \in L^1$ is an approximate identity if

$$\int_{\mathbb{R}^N} \varphi dx = 1$$

Then for $\varphi_h := \left(\frac{1}{h^N}\right)\varphi\left(\frac{x}{h}\right)$ we can say that:

- For $f \in L^p$, $f * \varphi_h \rightarrow f$ in L^p as $h \rightarrow 0$
- If f is continuous at x_0 , then $f * \varphi_h(x_0) \rightarrow f(x_0)$ as $h \rightarrow 0$
- φ is bounded above by $\frac{C}{1+|x|^{N+\alpha}}$, for $C, \alpha > 0$

Definition 4: Fourier Transform

The Fourier Transform of a function $f \in L^1$, denoted by $\hat{f}$ is defined as

$$\hat{f} = \int_{\mathbb{R}^N} f(x)e^{-2\pi i x \cdot t} dx$$

Definition 5: Riemann-Lebesgue Lemma

$\hat{f} \in C_0(\mathbb{R}^N) := \{f \in C(\mathbb{R}^N) : \lim_{|x| \rightarrow \infty} f(x) = 0\}$ for all $f \in L^1$, meaning that $\hat{f}$ is continuous and $\hat{f} \rightarrow 0$ as $|t| \rightarrow \infty$

Definition 6: Gaussian identity for the Fourier Transform

Let $\varphi(x) = e^{-\pi|x|^2}$ be the fixed point such that $\hat{\varphi} = \varphi$. More generically, we can then define a function $\psi_c(x) = e^{-c|x|^2}$ and its Fourier Transform as,

$$\hat{\psi}_c(t) = \left(\frac{\pi}{c}\right)^{\frac{N}{2}} \varphi\left(\sqrt{\frac{\pi}{c}}t\right)$$

Definition 7: Fourier Inversion Formula

The Fourier Inversion Formula for $f \in L^1$, relies on the convolution,

$$(f * \psi_\varepsilon)(x) = \int_{\mathbb{R}^N} \hat{f}(t)\varphi_\varepsilon(t)e^{2\pi i x t} dt$$

With a special case where $\varphi(t) = e^{-\pi|t|^2}$ and $\varphi_\varepsilon(t) = \frac{1}{\varepsilon^N}\varphi\left(\frac{t}{\varepsilon}\right)$ we can say that $(f * \psi_\varepsilon)(x) \rightarrow f$ If both $f, \hat{f} \in L^1$, then by the dominated convergence theorem, we can say that

$$f(x) = \int_{\mathbb{R}^N} \hat{f}(t) e^{2\pi i x t} dt$$